Project: Monte Carlo simulation of solid Ar

Objective: To create a Monte Carlo (MC) code for a Lennard-Jones (LJ) system in a solid phase. To analyse the influence of the temperature on the structure and thermodynamic properties.

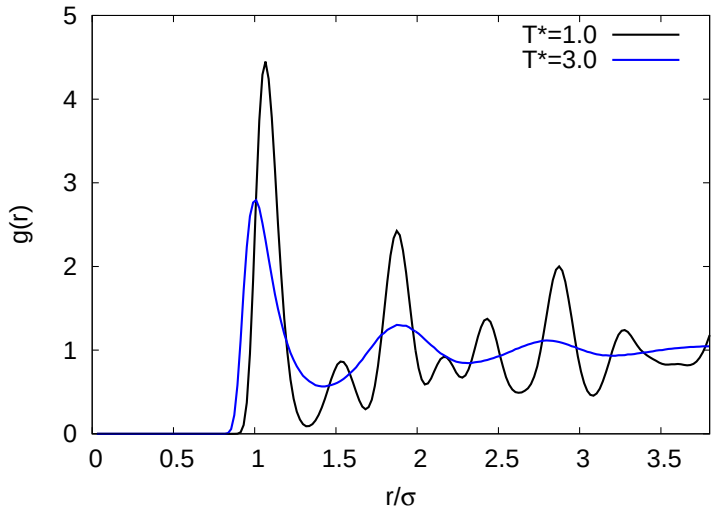
Outline

1. Write a MC code for a (N,V,T) simulation of a Lennard-Jones (LJ) system. Use N=500 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
2. Create an initial set-up of positions from the sites of a fcc (*face centred cubic*) lattice. (HINT: consider a fcc lattice as the superposition of 4 single cubic lattices.)
3. Run a MC simulation for argon at $T^* = 1.0$ and $\rho^* = 1.1$. Compute the interaction energy U^*, the pressure P^*, the radial distribution function $g(r)$ and the structure factor $S(k)$ of the system.
4. Repeat the simulation for the following temperatures: $T^* = 2.0$, $T^* = 3.0$ and $T^* = 4.0$, and compare with the previous results.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ar	39.948	119.8	3.405

Some results to compare with:



Radial distribution functions for different temperatures.

Questions

1. **Summarize the Metropolis algorithm for (N,V,T) simulations. Describe the main hypothesis involved in its derivation.**

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Describe how did you fix the Δ parameter for the trial moves.

4. Discuss separately the qualitative and quantitative differences you find in the $g(r)$ and $S(k)$ functions when heating the system.